

Abstract

UPI Transaction Analyzer

Unified Payments Interface (UPI) has become one of the most widely used digital payment systems in India, generating a large volume of transaction data every day. However, identifying customer spending behavior, transaction trends, and payment preferences from this data is difficult through manual analysis. This project analyzes the **UPI Transactions 2024** dataset, sourced from a publicly available Kaggle repository, containing approximately **250,000 synthetic yet realistic UPI transaction records** with attributes including transaction type, merchant category, transaction amount, transaction status, sender and receiver bank details, sender state, device type, network type, fraud indicators, and time-based information such as transaction hour, day of the week, and weekend indicators. Using **Python, Pandas, and NumPy**, the dataset is cleaned and preprocessed by handling missing values, removing duplicate records, validating data types, and processing timestamp information. Exploratory Data Analysis (EDA) using grouped aggregations, statistical summaries, and visualizations created with **Matplotlib and Seaborn** is performed to analyze customer spending across merchant categories, bank-wise transaction distribution, regional payment trends, transaction success rates, fraud occurrences, and time-based transaction patterns. The project aims to identify customer spending trends and transaction behavior, providing actionable insights that help merchants understand consumer preferences, optimize business strategies, and enable banks and payment service providers to improve transaction services and operational decision-making.

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